Area of the quadrilateral ABCD $= \frac{1}{2}\begin{vmatrix} x_1 & x_2 & x_3 & x_4 & x_1 \\ y_1 & y_2 & y_3 & y_4 & y_1 \end{vmatrix}$

$= \frac{1}{2}\{(x_1y_2 + x_2y_3 + x_3y_4 + x_4y_1) - (x_2y_1 + x_3y_2 + x_4y_3 + x_1y_4)\}$ sq.units.

Note

- To find the area of a quadrilateral, we divide it into triangular regions, which have no common area and then add the area of these regions.
- The area of the quadrilateral is never negative. That is, we always take the area of quadrilateral as positive.

Thinking Corner

1. If the area of a quadrilateral formed by the points (a, a), $(-a, a)$, $(a, -a)$ and $(-a, -a)$, where $a \neq 0$ is 64 square units, then identify the type of the quadrilateral
2. Find all possible values of a.

Example 5.1 Find the area of the triangle whose vertices are $(-3,5)$, $(5,6)$ and $(5,-2)$

Solution Plot the points in a rough diagram and take them in counter-clockwise order.

Let the vertices be $A(-3,5)$, $B(5,-2)$, $C(5,6)$

(x_1, y_1) (x_2, y_2) (x_3, y_3)

The area of ΔABC is

$= \frac{1}{2}\{(x_1y_2 + x_2y_3 + x_3y_1) - (x_2y_1 + x_3y_2 + x_1y_3)\}$

$= \frac{1}{2}\{(6 + 30 + 25) - (25 - 10 - 18)\}$

$= \frac{1}{2}\{61 + 3\}$

$= \frac{1}{2}(64) = 32$ sq.units

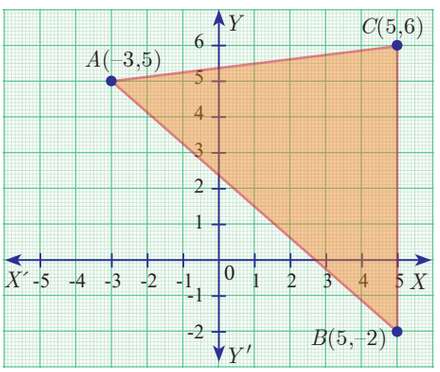


Fig. 5.10

Example 5.2 Show that the points $P(-1.5,3)$, $Q(6,-2)$, $R(-3,4)$ are collinear.

Solution The points are $P(-1.5,3)$, $Q(6,-2)$, $R(-3,4)$

Area of ΔPQR $= \frac{1}{2}\{(x_1y_2 + x_2y_3 + x_3y_1) - (x_2y_1 + x_3y_2 + x_1y_3)\}$

$= \frac{1}{2}\{(3 + 24 - 9) - (18 + 6 - 6)\} = \frac{1}{2}\{18 - 18\} = 0$

Therefore, the given points are collinear.

Example 5.3 If the area of the triangle formed by the vertices $A(-1,2)$, $B(k,-2)$ and $C(7,4)$ (taken in order) is 22 sq. units, find the value of k.

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